

Droop-core nonemptiness for approval committees with two omissions

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Abstract

Core stability requires that no coalition can use its proportional share of seats to make every member strictly better off. We prove that every finite approval election with $m \geq 3$ candidates and $k = m - 2$ seats has a deterministic Droop-core committee, and hence a Hare-core committee, without restricting the number of voters or their approval sets. The Droop rule can let smaller coalitions fund a challenge, making stability more demanding than under Hare. Individual improvements are strict, and equality at the Droop threshold does not block. The proof bounds repeated support for blocking targets by the size of their intersection. This estimate justifies a contraction that fixes an initial blocking target, removes its strict gainers and leaves a smaller ordinary election with two omissions. Every Droop core of the residual election lifts to a Droop core of the original election. Induction gives a finite exact construction, with overlapping deviations and an explicit zero-seat boundary. Together with a one-swap bound, this establishes Droop-core existence whenever at most two candidates are unselected.

1 Introduction

A representative committee can be challenged by voters who would all prefer to use their collective entitlement to select a different set. The Hare core rules out every such challenge when a coalition's share of seats is proportional to its share of the electorate. With binary approvals, each voter's utility is the number of approved candidates selected. The resulting stability requirement is stronger than a comparison of total approval scores: each member of a blocking coalition must strictly improve.

Proportional Approval Voting (PAV) assigns diminishing harmonic credit for each additional approved representative. Peters' one-swap bound gives core existence when $k = m$ or $k = m - 1$ [1, Lemma 3.1(ii)]. He also proved existence for $k \leq 8$ and, using a recursive PAV procedure, for $m \leq 15$ [1, Theorems 4.1, 4.5 and 5.3]. His procedure fixes blocking targets and removes the voters who strictly gain from them before reoptimizing over full committees [1, Algorithm 1]. Becker, Greger and Peters establish existence with at most seven distinct voter types by a computer-assisted argument [2, Theorems 2.5 and 5.2]. The result here extends the omission-based guarantee to $m - k = 2$ under the stronger Droop criterion, with no bound on candidate count or voter types. There is no restriction on how many candidates a voter approves or disapproves. This quota distinction matters: Peters shows that PAV can fail the Droop core already at $k = 6$ [1, Example 6.1].

The main step is a contraction theorem rather than a claim that PAV itself always selects a core committee. Start from a global PAV maximizer and a blocking target F . If some of its strict gainers can strictly gain again after F is selected, two omissions force their approval sets into a specific configuration. A summed swap inequality then bounds their number in terms of overlap with the later target. This bound permits a new ordinary residual election on $C \setminus F$, with fewer voters and $k - |F|$ seats. Every residual core lifts, so no additional invariant or tie-breaking condition is required.

The proof is analytic. We state the Droop result first, then develop the contraction through a direct Hare proof before proving the stronger statement. Exact examples illustrate the distinction between the quotas, the contraction and its repeated-support bound; the computations are not premises of the proof.

2 Model and main theorem

Let N be a nonempty finite voter set, $n = |N|$, and let C be a finite candidate set with $m = |C|$. Voter i approves an arbitrary set $A_i \subseteq C$. For every $X \subseteq C$, put $u_i(X) = |A_i \cap X|$. A committee has exactly k candidates, where $1 \leq k \leq m$. A nonempty coalition $S \subseteq N$ Hare-blocks a committee W by a target $T \subseteq C$ if

$$n|T| \leq k|S| \quad \text{and} \quad u_i(T) > u_i(W) \quad \text{for every } i \in S.$$

An empty target cannot strictly improve a voter, and affordability implies $|T| \leq k$. Define the full strict-gainer set and its size by

$$Q(W, T) = \{i \in N : u_i(T) > u_i(W)\}, \quad q(W, T) = |Q(W, T)|.$$

Every blocking coalition is contained in $Q(W, T)$. Conversely, if a nonempty target satisfies $kq(W, T) \geq n|T|$, its full strict-gainer set is nonempty and is itself an affordable coalition. Thus the exact core test is

$$W \text{ is Hare core} \iff kq(W, T) < n|T| \quad \text{for every } \emptyset \neq T \subseteq C, |T| \leq k. \quad (1)$$

Equality is blocking under this Hare convention. Under the Droop convention of Peters [1, Section 6], affordability instead requires $n|T| < (k+1)|S|$. Taking the full strict-gainer set again shows that a nonempty target Droop-blocks W exactly when

$$(k+1)q(W, T) > n|T|. \quad (2)$$

A Droop-core committee admits no such target. Individual improvement remains strict, but equality in (2) does not block. Since $q(W, T) \leq n$, an affordable integral target has size at most k . Every Droop-core committee is Hare core: for a nonempty target, $kq \geq n|T|$ implies $q > 0$ and hence $(k+1)q > n|T|$. Neither convention requires unanimous approval of a target, and targets may overlap the committee. Unqualified uses of *core* and *blocker* in Sections 3–5 refer to Hare, unless both conventions are explicitly mentioned.

Theorem 1. *For every finite approval election with $m \geq 3$ candidates and $k = m - 2$ seats, a deterministic Droop-core committee exists. In particular, the ordinary deterministic Hare core is nonempty.*

The Droop statement is proved in Section 6.

Proportional Approval Voting (PAV) maximizes a harmonic approval score. We use its optimality only to initialize a contraction; a PAV maximizer need not itself be core. Define

$$H_0 = 0, \quad H_a = \sum_{j=1}^a \frac{1}{j} \quad (a \geq 1), \quad \text{PAV}(X) = \sum_{i \in N} H_{u_i(X)}.$$

A global PAV maximizer among k -committees exists by finiteness. It is also a one-swap local maximizer: replacing any selected candidate by any unselected candidate does not increase PAV.

3 Bounding repeated support

We first record an elementary removal bound. Fix a committee W , an unselected candidate b , and a nonempty set $Y \subseteq W$. For one voter, sum the PAV changes over the swaps $W - \{y\} + \{b\}$, $y \in Y$. If b is approved, every such change is nonnegative. If b is not approved and $u = u_i(W) > 0$, the sum is $-|A_i \cap Y|/u \geq -1$. If b is not approved and $u = 0$, the sum is zero. Thus this sum is at least -1 for every voter; no division by zero is used.

The following numerical bound implies the one-outsider exclusion of Peters [1, Lemma 3.1(ii)] under both quota conventions.

Lemma 2 (One-outsider bound). *Let W be a one-swap PAV local maximizer with $|W| = k$. For every nonempty $T \subseteq C$ with $|T| \leq k$ and $|T \setminus W| \leq 1$,*

$$(k+1)q(W, T) \leq n|T|. \quad (3)$$

Thus such a target is neither a Hare blocker nor a Droop blocker. If exactly two candidates are unselected, every blocker under either convention contains both of them.

Proof. If $T \subseteq W$, then $q(W, T) = 0$, so (3) holds. Otherwise write $T \setminus W = \{b\}$. Write $t = |T|$ and $Y = W \setminus T$, so $d = |Y| = k - t + 1 \geq 1$. A strict gainer approves b and none of Y . Its utility at W is at most $t - 1$; hence each of the d swaps adding b and removing a member of Y increases its PAV term by at least $1/t$. Every other voter's summed contribution is at least -1 by the removal bound. Writing $q = q(W, T)$, the total change, summed over all d swaps and voters, is therefore at least

$$\frac{dq}{t} - (n - q) = \frac{(k+1)q - nt}{t}.$$

Local maximality makes each swap's total change nonpositive, so $(k+1)q \leq nt$. This excludes Droop blocking directly. If $q > 0$, it also gives $kq < (k+1)q \leq nt$; if $q = 0$, then $kq = 0 < nt$. Thus Hare blocking is also excluded, proving the consequences. $\square$

The next estimate holds for *every* completion of a target containing both initially unselected candidates. Neither affordability of that target nor optimality of the later committee is needed.

Lemma 3 (Repeated-support bound). *Let $|C| = k+2$, let W_0 be a one-swap PAV local maximizer, and let $C \setminus W_0 \subseteq F \subseteq C$ with $|F| \leq k$. Put $Q_0 = Q(W_0, F)$. For any k -committee $W \supseteq F$ and any $T \subseteq C$ with $|T| \leq k$, define*

$$I = F \cap T, \quad s = |I|, \quad R = Q_0 \cap Q(W, T), \quad r = |R|.$$

Then

$$(k+1)r \leq ns. \quad (4)$$

The later target T is not required to be affordable.

Proof. If $r = 0$, the conclusion is immediate. Suppose $r > 0$ and put $B = C \setminus W_0$, so $|B| = 2$. For $i \in R$, let $a = |A_i|$. The two strict improvements and the containment $F \subseteq W$ give

$$a \geq u_i(T) \geq u_i(W) + 1 \geq u_i(F) + 1 \geq u_i(W_0) + 2 \geq a.$$

Every inequality is therefore an equality. In particular, the voter approves both members of B , F contains exactly $a - 1$ of its approvals, and T contains all its approvals. By assumption $B \subseteq F$, so $B \subseteq I$. The unique approved candidate d_i outside F lies in $W_0 \setminus I$, and every other approved candidate lies in I . Consequently $a \geq 3$ and $a - 1 \leq s$, in particular $s \geq 2$.

Set $Y = W_0 \setminus I$. Since both unselected candidates belong to I ,

$$d = |Y| = k - s + 2.$$

Sum the PAV changes over all $2d$ swaps adding $b \in B$ and removing $y \in Y$. For a voter $i \in R$, both additions are approved, exactly one member of Y is approved, and $u_i(W_0) = a - 2$. This voter's summed contribution is exactly

$$\frac{2(d-1)}{a-1} \geq \frac{2(k-s+1)}{s}.$$

For each voter outside R , the removal bound applied separately to the two choices of b gives a summed contribution of at least -2 . Every swap has nonpositive aggregate change, so

$$0 \geq \frac{2r(k-s+1)}{s} - 2(n-r).$$

Multiplication by the positive number $s/2$ and rearrangement yield $(k+1)r \leq ns$, as required. $\square$

4 A core-preserving contraction

Lemma 4 (Contraction and lifting). *Let $|C| = k + 2$, let W_0 be a global PAV maximizer, and let F block W_0 . Write*

$$f = |F|, \quad Q_0 = Q(W_0, F), \quad q_0 = |Q_0|, \quad D = C \setminus F, \quad N' = N \setminus Q_0, \\ k' = k - f, \quad n' = n - q_0.$$

Then $k' \geq 1$, $n' \geq 1$, $|D| = k' + 2$, and $kn' \leq nk'$. Consider the ordinary residual approval election with electorate N' , candidates D , approval sets $A_i \cap D$, and k' seats. Every core committee U of this residual election lifts to an original core committee $F \cup U$.

Proof. First, $q_0 < n$: otherwise any padding of F to k candidates would strictly improve every voter over W_0 , and hence strictly increase PAV, contrary to global maximality. The original blocking inequality $kq_0 \geq nf$ then also implies $f < k$. Thus $n', k' > 0$ and

$$kn' = k(n - q_0) \leq n(k - f) = nk'. \quad (5)$$

The candidate count satisfies $|D| = k' + 2$.

Let U be a residual core committee and set $W = F \cup U$. Suppose a nonempty target T of size $t \leq k$ blocks W in the *original* election, so $kq \geq nt$ for $q = q(W, T)$. Define

$$I = F \cap T, \quad s = |I|, \quad V = T \setminus F, \quad v = |V|, \\ r = |Q_0 \cap Q(W, T)|, \quad q' = q - r.$$

Lemma 2 gives $C \setminus W_0 \subseteq F$, so Lemma 3 gives $(k+1)r \leq ns$, hence $kr \leq ns$. It follows that

$$kq' = kq - kr \geq n(t-s) = nv. \quad (6)$$

The target V is nonempty, since $T \subseteq F \subseteq W$ would make strict improvement impossible. Every voter $i \in Q(W, T) \setminus Q_0$ strictly improves V over U in the residual election: cancellation of the fixed part gives

$$|A_i \cap V| > |A_i \cap U| + |A_i \cap (F \setminus T)| \geq |A_i \cap U|.$$

Combining (5) and (6),

$$kk'q' \geq nk'v \geq kn'v, \quad \text{so} \quad k'q' \geq n'v.$$

As $q' \leq n'$ and $n' > 0$, this inequality implies $v \leq k'$. Thus V is a nonempty admissible target and the q' remaining voters form an affordable strictly improving residual coalition, contradicting the core property of U . Any additional residual strict gainers only strengthen this contradiction. $\square$

The residual election is a separately defined induction instance; its quota is not substituted into an original blocking test. Every original test retains the original n, k . Equations (5)–(6) prove exactly why a hypothetical original blocker would induce a residual blocker.

This contraction differs from fixed-target PAV reoptimization. With F , N' and $U \subseteq D$ as above, the two objectives are

$$\underbrace{\sum_{i \in N'} H_{|A_i \cap F| + |A_i \cap U|}}_{\text{full-committee score with } F \text{ fixed}} \quad \text{and} \quad \underbrace{\sum_{i \in N'} H_{|A_i \cap U|}}_{\text{ordinary residual-election score}}.$$

Peters' recursive PAV rule uses the first objective [1, Algorithm 1]; our recursive initialization uses the second. They do not generally differ by a constant independent of U . More importantly, Lemma 4 accepts every residual core, irrespective of how it is obtained.

5 Induction and exact construction

We first prove the Hare-core consequence of Theorem 1 directly; Section 6 establishes its stronger Droop statement.

Direct proof of Hare-core nonemptiness. Induct on $k \geq 1$, always with $|C| = k + 2$. For $k = 1$, choose a global PAV maximizer. Any blocker would have size one and $q = n$, hence strictly increase PAV, a contradiction.

For $k > 1$, choose a global PAV maximizer W_0 . If it is core, the claim is proved. Otherwise choose any blocking target F . Lemma 4 produces a residual ordinary approval election with a nonempty electorate, $k' + 2$ candidates and $1 \leq k' < k$ seats. The induction hypothesis gives a residual core committee U . Lemma 4 lifts it to the original core committee $F \cup U$, which has exactly $f + k' = k$ candidates. $\square$

The proof gives the following exact construction.

HareCore $(N, C, k, (A_i)_{i \in N})$, where $|C| = k + 2$.

1. Choose a global PAV maximizer W_0 among the k -committees.
2. If no target blocks W_0 under (1), return W_0 . Otherwise choose any blocker F .
3. Set $N' = N \setminus Q(W_0, F)$, $D = C \setminus F$ and $k' = k - |F|$.
4. Compute $U = \mathbf{HareCore}(N', D, k', (A_i \cap D)_{i \in N'})$ and return $F \cup U$.

The positive integer seat count strictly decreases at each recursive call. Any tied PAV maximizer and any first blocker are allowed. At a call with m candidates, PAV maximization compares $\binom{m}{2}$ committees; the exhaustive blocking test examines $\sum_{t=1}^{m-2} \binom{m}{t} = 2^m - m - 2$ targets. Thus the reference implementation is not a polynomial-time algorithm.

All lemmas sum over actual voters, so repeated types with positive integer multiplicities are included. Empty approvals make zero PAV changes and never strictly gain; universal approvals cannot strictly gain from a target of at most k candidates. These voters are nevertheless retained in every electorate unless they genuinely belong to a deleted full-gainer set. Finiteness is used for global maximization, exhaustive tests and the well-founded induction.

5.1 Contraction of a blocked PAV maximizer

Let $C = B \dot{\cup} X \dot{\cup} Y$, where $B = \{b_0, b_1\}$, $|X| = 3$ and $|Y| = 6$. There are 18 voters: five approve $X \cup \{b_0\}$, five approve $X \cup \{b_1\}$, and eight approve Y . Set $k = 9$.

The committee $W_0 = X \cup Y$ is the unique global PAV maximizer. To see this, measure the PAV loss of a committee relative to the full candidate set C . Every committee omits a pair, and its loss depends only on the pair's classes:

omitted-pair classes	BB	BX	BY	XX	XY	YY
$\text{PAV}(C) - \text{PAV}(W)$	$\frac{5}{2}$	$\frac{25}{6}$	$\frac{31}{12}$	$\frac{35}{6}$	$\frac{23}{6}$	$\frac{44}{15}$

The BB loss is strictly smallest. Nevertheless, $F = X \cup B$ blocks W_0 : the first ten voters improve from utility three to four, and

$$kq(W_0, F) = 9 \cdot 10 = 18 \cdot 5 = n|F|.$$

For $k = 9$, this example has fewer voters than the 27-voter unique-global-PAV counterexample described as voter-minimal in Peters [1, Section 4.3]. We make no minimality claim for eighteen voters. This comparison is specific to $k = 9$: a PAV failure with $k = 18$ and three voters appears in Becker, Greger and Peters [2, Figure 1].

After contraction, eight voters remain, all approving the six candidates in Y , and the residual committee size is four. Every four-element subset $U \subseteq Y$ is residual core because no target of at most four candidates can strictly improve any voter. Lemma 4 therefore makes $F \cup U$ an original core committee. The accompanying code reproduces this construction and independently checks every original target, using integers for all quota tests. Here the deleted voters approve no candidate outside F , so repeated support is zero. For a nonzero example, keep B, X, Y, k but change the three multiplicities to 27, 27, 44 and add one voter approving $B \cup \{y_1\}$, where $y_1 \in Y$. The resulting 99-voter profile again has the unique PAV maximizer $W_0 = X \cup Y$ and the Hare blocker $F = B \cup X$, now with $9 \cdot 55 = 99 \cdot 5$. Choose any four-element $U \subseteq Y$ excluding y_1 . For the added voter,

$$u_i(W_0) = 1, \quad u_i(F) = u_i(F \cup U) = 2, \quad u_i(B \cup \{y_1\}) = 3.$$

Thus a deleted voter can strictly gain again. The later target $T = B \cup \{y_1\}$ has $q(F \cup U, T) = 1$ and is not affordable under either quota: $9 \cdot 1 < 99 \cdot 3$ and $10 \cdot 1 < 99 \cdot 3$. The example illustrates the scope of Lemma 3, not a blocker of the lifted committee. Its validity does not require deleted voters to be permanently unable to gain. The accompanying checker replays the bound for all completions of F and all legal targets.

6 The Droop strengthening

We now use the strict Droop test (2) throughout the induction.

Proof of Theorem 1. For induction only, also allow $k = 0$ and $m = 2$, with a nonempty electorate. The empty committee is then Droop core: no nonempty target can satisfy (2). This is an auxiliary base case, not a change to the positive-seat model of Section 2.

Let $k \geq 1$ and choose a global PAV maximizer W_0 . If it is Droop core, stop. Otherwise choose a Droop blocker F , and put $f = |F|$, $Q_0 = Q(W_0, F)$ and $q_0 = |Q_0|$. Lemma 2 implies that F contains both candidates outside W_0 , and Lemma 3 applies. Moreover $q_0 < n$, since otherwise padding F to k candidates strictly increases every voter's PAV term.

Define $D = C \setminus F$, $N' = N \setminus Q_0$, $k' = k - f$ and $n' = n - q_0$. These satisfy $n' > 0$, $0 \leq k' < k$, $|D| = k' + 2$, and

$$(k + 1)n' < n(k' + 1). \quad (7)$$

By induction the ordinary residual election on N', D with projected approvals and k' seats has a Droop core U . We show that every such U lifts to a Droop core $W = F \cup U$.

Suppose instead that T Droop-blocks W . Write $q = q(W, T)$, $t = |T|$, $s = |F \cap T|$, $V = T \setminus F$, $v = |V|$, $r = |Q_0 \cap Q(W, T)|$ and $q' = q - r$. Since $T \subseteq F$ cannot strictly improve anyone over W , we have $v \geq 1$. Lemma 3 and (2) give

$$(k + 1)q' = (k + 1)(q - r) > n(t - s) = nv.$$

Using (7) and only positive integer multipliers,

$$(k + 1)(k' + 1)q' > n(k' + 1)v > (k + 1)n'v, \quad \text{hence} \quad (k' + 1)q' > n'v.$$

The q' voters outside Q_0 strictly improve from U to V after cancelling the fixed part, exactly as in Lemma 4. Also $q' \leq n'$ and $n' > 0$, so $v < k' + 1$. Integrality gives $v \leq k'$. For $k' = 0$ this contradicts $v \geq 1$; for $k' > 0$ it makes V an admissible Droop blocker of U , again a contradiction. The lift has exactly $f + k' = k$ candidates, completing the induction. $\square$

The same exhaustive construction therefore works with the Droop test (2) and an empty-committee return at the auxiliary zero-seat base case. The original Hare construction and its equality convention remain as stated in Section 5.

Corollary 5. *Every finite approval election with $1 \leq k \leq m$ and $m - k \leq 2$ has a deterministic Droop-core committee, and hence a Hare-core committee.*

Proof. For $m - k \leq 1$, choose a global PAV maximizer W . Every legal target has $|T \setminus W| \leq 1$, so Lemma 2 excludes Droop blocking. The case $m - k = 2$ is Theorem 1. $\square$

6.1 A Hare core outside the Droop core

Peters' Example 6.1 [1] separates the two quota conventions within the two-omission class. Relabel its candidates as $C = B \dot{\cup} X \dot{\cup} Y$, with $B = \{b_0, b_1\}$, $|X| = 2$, $|Y| = 4$ and $k = 6$. The 24 voters have the following approvals:

approval set	multiplicity
$X \cup \{b_0\}$	7
$X \cup \{b_1\}$	7
Y	10

The unique global PAV maximizer is $W_0 = X \cup Y$. For omitted-pair classes BB , BX , BY , XX , XY and YY , the respective losses relative to $\text{PAV}(C)$ are $14/3, 49/6, 29/6, 35/3, 43/6, 35/6$; the BB loss is uniquely smallest. For $F = X \cup B$, both seven-voter types strictly improve, and

$$6 \cdot 14 = 84 < 24 \cdot 4 = 96 < 7 \cdot 14 = 98.$$

Thus F Droop-blocks W_0 but does not Hare-block it. In fact, W_0 is Hare core. The ten voters approving Y are satiated. A target improving exactly one seven-voter type needs at least three candidates; improving both requires all four candidates of F . The Hare tests fail in both cases, since $6 \cdot 7 < 24 \cdot 3$ and $6 \cdot 14 < 24 \cdot 4$.

Consequently the Hare construction returns W_0 , whereas the Droop construction fixes F and leaves ten voters, four candidates and two seats. Every pair $U \subseteq Y$ is a residual Droop core and lifts to $F \cup U$. The supplied input and independent verifier reproduce the two outputs and check all 246 original legal targets for each.

6.2 Why Droop equality must remain nonblocking

The strict inequality in (2) cannot be replaced by a weak inequality in the existence theorem. For any $k \geq 1$, take $n = k + 1$ voters and $C = \{c_1, \dots, c_{k+1}, d\}$. Voter i approves only c_i , and no voter approves d . Every k -committee W omits some c_i . For $T = \{c_i\}$, only voter i strictly gains, so

$$(k + 1)q(W, T) = k + 1 = n|T|.$$

Hence the modified convention that blocks at Droop equality has an empty core even with two omissions. This is a boundary of the quota convention, not an empty-core example under (2) or a tightness claim for the repeated-support bound.

7 Discussion

The two-omission condition is used at one precise point. A voter who strictly improves first to F and then to a target against a completion of F must start exactly two approvals below its total and gain one approval at each step. This rigidity turns repeated participation into the overlap bound of Lemma 3. Counting two possible gains per voter alone would not justify deleting a coalition. The overlap bound supplies the missing charge for candidates cancelled from a later target.

With three or more omissions, two strict improvements no longer force the utility equalities used in Lemma 3. The proof therefore does not establish core existence for unrestricted (m, k) . Its conclusion is a uniform integral existence theorem for the entire two-omission class, under both Hare and Droop affordability, supported by contraction and lifting.

AI use disclosure

OpenAI Codex, using GPT-6 Astra, was used for proof exploration and checking, exact-verifier development, and manuscript drafting and revision. AI-generated output is not treated as mathematical evidence. The results are established by the analytic proofs presented above. The accompanying code implements the construction and independently checks its outputs and the illustrative examples. The author remains responsible for the statements, proofs, citations, source code, and final artifacts.

Code availability

The constructor and verifier source code, example instances, manuscript source, and reproduction instructions accompany this manuscript. The project repository is <https://github.com/Baymax-ray/approval-core-two-omissions>.

References

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